

School of Business, Law & Entrepreneurship



Unit Outline

This content is protected and may not be shared, uploaded, or distributed.

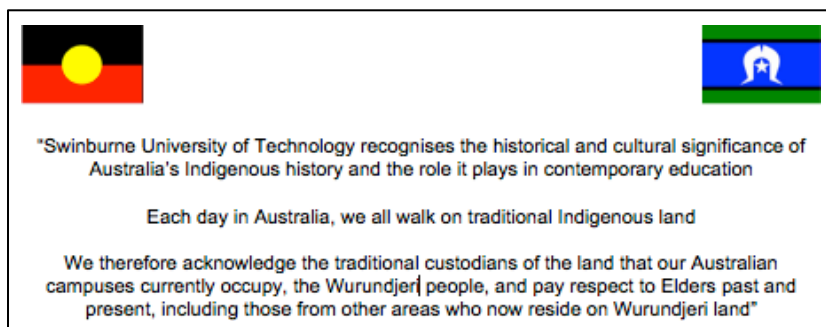
INF10025

Data Management and Analytics

Semester September 2026

Please read this Unit Outline carefully. It includes:

- PART A** Unit summary
- PART B** Your Unit in more detail
- PART C** Further information



PART A: Unit Summary

Unit Code(s)	INF10025
Unit Title	Data Management and Analytics
Duration	12 Weeks
Total Contact Hours	36 Hours
Requisites:	
Pre-requisites	Nil
Co-requisites	Nil
Concurrent pre-requisites	Nil
Anti-requisites	INF20004 Database Concepts and Modelling
Assumed knowledge	Competent use of an operating system such as Windows or Mac OSX. Competent use of a Web Browser. Competent use of a file management application such as Windows File Explorer or OSX Finder. Competent use of a text editor application such as Notepad. Note: Non Windows users (such as Mac OSX) will need to use Windows Software. The easiest way to do this is via the Azure Virtual Desktop portal
Credit Points	12.5
Campus/Location	Swinburne Vietnam
Mode of Delivery	Face-to-face, 2 hour Lab, 1 hour asynchronous Lecture, and asynchronous study
Assessment Summary	Portfolio

Aims

This unit will provide a solid foundation for the design, implementation and management of organisational databases. Organisational data and data modelling is introduced, focusing on structured and unstructured data, and entity-relationship (ER) modelling. The skills required to construct ER diagrams will be taught, with a focus on ensuring that the logic of the model reflects the real-world industry case it is representing. Relational databases will be introduced and the functionality they afford organisations will be explored through real world industry cases. The process of designing, building and retrieving information from a database using SQL will be a focus of this unit. The unit also introduces students to the role databases play in data analytics and helping organisations harness the power of and insights from their data.

Unit Learning Outcomes

Students who successfully complete this unit can:

ULO1	Define and explain fundamental data and database concepts including tables, relations, keys, queries, transactions, and structured, semi-structured, unstructured data
ULO2	Create, store, retrieve, exploit, and visualise data using modern database tools, functions, and techniques
ULO3	Understand and apply conventional data modelling techniques to solve practical database design problems
ULO4	Explain the impact of big data and modern data analytics tools in the design, use, and management of database systems
ULO5	Examine the application of database technologies and methods in Blockchain, Artificial Intelligence, and Chatbot technologies and social media

Graduate Attributes

The Swinburne Graduate Attributes describe the capability of our graduates to use knowledge, skills and behaviours to contribute to society meaningfully and positively. They include professional, self-directed learning and future-ready skills.

This unit may contribute to the development of the following Swinburne Graduate Attributes:

- Communication 1 - Verbal communication
- Communication 2 - Communicating using different media
- Teamwork 1 - Collaboration and negotiation
- Teamwork 2 - Teamwork roles and processes
- Digital literacies 1 – Information literacy
- Digital Literacies 2 – Technical literacy

Other graduate attributes may be practised in the unit but are not formally taught as part of the unit content, nor incorporated within formal assessment.

Content

- Structured and unstructured data
- Database relations, tables, records, keys, filters, queries, parameters, forms, indexes
- Data formats, manipulation, and exchange tools
- Transactions and ACID
- Entity-relationship diagrams and modelling
- SQL
- Data warehousing, big data, and data normalisation
- Data analytics, business intelligence, data visualisation

PART B: Your Unit in more detail

Unit Improvements

Feedback provided by previous students through the Student Survey has resulted in improvements that have been made to this unit. Recent improvements include:

- Standardised Canvas Template

Unit Teaching Staff

Name	Role	Email	Consultation Times
(HN) Thu Pham	Unit Coordinator	thupham@swin.edu.au	By appointment via email

Category	Activity	Total Hours	Hours per Week	Teaching Period Weeks
Online	Lectures	12 hours	1 hours	Weeks 1 to 12
In person	Classes / Lab	24 hours	2 hours	Weeks 1 to 12

Week by Week Schedule

Week	Week Beginning	Teaching and Learning Activity	Student Task or Assessment	
1	14 Sep	Fundamental DBMS Concepts and Tools	- Review weekly lectures and supplementary materials - Complete weekly readings and tasks - Update Learning Summary Report - Submit weekly tasks for feedback and sign off.	
2	21 Sep	Relationships, Indexes, and Queries		
3	28 Sep	Queries and Data Import/Export		Task 1 submit
4	05 Oct	Business Intelligence Tools		
5	12 Oct	SQL and Modelling Part 1		Test 1 in tutorial
6	19 Oct	SQL and Modelling Part 2		Task 2 submit
7	26 Oct	SQL and Modelling Part 3		
8	02 Nov	SQL and Modelling Part 4		Task 3 submit Test 2 in tutorial
9	09 Nov	Static Data, Variable Data, More Modelling, and Views		Test 3 in tutorial
10	16 Nov	Normalisation, NoSQL, and JSON		Task 4 Submit
11	23 Nov	Transactions, ACID, ETL, and Data Warehousing		D Task submit
12	30 Nov	Summary and Review		Submit: • LSR • Task 5 • HD Tasks

Assessment

a) Assessment overview

Tasks and Details	Individual or Group	Weighting	Unit Learning Outcomes that this assessment task relates to	Assessment Due Date
Learning Portfolio consisting of: • Tests • Pass, Credit, Distinction, and / or High Distinction Tasks • Learning Summary Report (LSR)	Individual	100%	All	Observe submission deadlines Tests are to be done during designated time slots (weeks 5, 8 & 9) Submission of LSR is end of week 12

Assessment Requirements	Details
Use of generative AI (genAI) in this unit	The valid use of genAI in this unit is as follows: The assessments and tests in this unit should be completed entirely without genAI assistance <u>in any way</u>. Ai tools and other various techniques will be used to discover assessments that have used genAI.
Hurdle requirements	- Test 1 and Test 2 must be satisfactorily completed - A minimum of four Pass Level tasks must be completed satisfactorily.
Final assessment period	There is no final exam in this unit
Submission requirements	Assignments and other assessments are generally submitted online through the Canvas assessment submission system which integrates with the Turnitin . Please ensure you keep a copy of all assessments that are submitted.
Extensions, late penalties and special consideration	<u>Extensions</u> If you need an extension you may apply for an automatic 3 day extension to the due date. Provided the assessment due date has not passed, you can use the EzExtend tool to apply for an extension, which is available on the left side of Canvas under “Assignment Extensions”. EzExtend lists assessment tasks 14 days before the due date. If you need an extension longer than 3 days, your assignment is already overdue, or you are experiencing exceptional circumstances, please consider applying for Special Consideration. Please note that EzExtend cannot be used to request individual extensions for time limited (e.g. exams, tests), synchronous performance or practice-based assessments requiring validation, synchronous oral assessments or group assessments. Assessment tasks that utilise tools outside of the LMS may require requests for extensions via the Special Consideration process. <u>Late Penalties</u> - Unless an extension has been approved, late submissions are not accepted in this unit . This is due to work being graded “Complete” or “Incomplete”, there is no way to make a 5% reduction in marks.

	<u>Special Consideration</u> If your studies have been adversely affected by serious and unavoidable circumstances outside of your control (e.g. severe illness or unavoidable obligation), you can apply for special consideration. Applications for Special Consideration are made via an online form <u>no later than 5.00pm</u> of the third working day after the assessment due date. For further information about eligibility and how to apply, see https://www.swinburne.edu.au/life-at-swinburne/student-support-services/special-consideration-assistance/
• Referencing	To avoid breaching academic integrity, you are required to provide references whenever you include information from other sources in your work and acknowledge when you have used Artificial Intelligence (AI) tools (such as ChatGPT). Further details regarding academic integrity are available in Section C of this document. Referencing conventions required for this unit are: Harvard Standard Helpful information on referencing can be found at http://www.swinburne.edu.au/library/referencing/
• Groupwork guidelines	There is no groupwork assessments in this unit. All assessments must be completed individually. Any collaboration between students on individual assessments constitutes Academic Misconduct.

b) Final unit results

a) Supplementary assessment	Students will be notified by the University if they are eligible for supplementary assessment. Students are offered supplementary assessment in this unit where they have: • failed the unit and the final mark for the unit is between 45 and 49, and • attempted all assessment tasks during the teaching and final assessment periods. Students are not eligible for supplementary assessment if they have received a mark reduction as a result of an academic misconduct finding under the <i>Student Academic Misconduct</i> regulations. Students may only undertake supplementary assessment in one unit per teaching period. Where students are eligible for supplementary assessment in
---	---

	multiple units in a teaching period, they may select which unit they will undertake supplementary assessment in. If students pass the supplementary assessment task, their final result for the unit will be changed to a Passed Supplementary (SP) with a mark of 50%. Further information on supplementary assessments is available on the Swinburne website.
--	--

Required Textbook(s)

There are no required textbooks in this unit

Recommended Reading Materials

The Library has a large collection of resource materials, both texts and current journals. Listed below are some references that will provide valuable supplementary information to this unit. It is also recommended that you explore other sources to broaden your understanding.

- Lecture slides can be downloaded from the Canvas Modules section. These include details on the material you will need to read each week, as well as tasks for you to undertake.
- Lecture recordings can be download from Canvas Modules section.
- Tasks handouts also contain a **list of references** that can be used.

PART C: FURTHER INFORMATION



For further information on any of these topics, refer to Swinburne's Student webpage <http://www.swinburne.edu.au/student/>

Student behaviour and wellbeing

All students are expected to: act with integrity, honesty and fairness; be inclusive, ethical and respectful of others; and appropriately use University resources, information, equipment and facilities. All students are expected to contribute to creating a work and study environment that is safe and free from bullying, violence, discrimination, sexual harassment, vilification and other forms of unacceptable behaviour.

The [Student Charter](#) describes what students can reasonably expect from Swinburne in order to enjoy a quality learning experience. The Charter also sets out what is expected of students with regards to your studies and the way you conduct yourself towards other people and property.

You are expected to familiarise yourself with University regulations and policies and are obliged to abide by these, including the [Student Academic Misconduct Regulations](#), [Student General Misconduct Regulations](#) and the [People, Culture and Integrity Policy](#). Any student found to be in breach of these may be subject to disciplinary processes.

Examples of expected behaviours are:

- conducting yourself in teaching areas in a manner that is professional and not disruptive to others
- following specific safety procedures in Swinburne laboratories, such as wearing appropriate footwear and safety equipment, not acting in a manner which is dangerous or disruptive (e.g. playing computer games), and not bringing in food or drink
- following emergency and evacuation procedures and following instructions given by staff/wardens in an emergency response.

Canvas

You should regularly log on to the Swinburne learning management system, Canvas. You can access Canvas via the [Student login](#) webpage or <https://swinburne.instructure.com/> Canvas is updated regularly with important unit information and communications.

Communication

All communication will be via your Swinburne email address. If you access your email through a provider other than Swinburne, then it is your responsibility to ensure that your Swinburne email is redirected to your private email address.

Academic Integrity

Academic integrity is about taking responsibility for your learning and submitting work that is honestly your own. It means acknowledging the ideas, contributions and work of others; referencing your sources and acknowledging the use of generative artificial intelligence;

contributing fairly to group work; and completing tasks, tests and exams without cheating. Artificial intelligence tools should only be used where approved by the Unit Convenor.

Swinburne University uses the Turnitin system, which helps to identify inadequate citations, poor paraphrasing and unoriginal work in assignments that are submitted via Canvas. Your Unit Convenor will provide further details.

Plagiarism, collusion, contract cheating, unauthorised file sharing, falsification, fabrication, manipulation or misrepresentation of information, reuse of previous work and non-compliance with instructions in an invigilated or non-invigilated assessment item are all breaches of academic integrity and treated as academic misconduct. Examples of breaches of academic integrity include, but are not limited to:

- submitting work as your own for assessment that has been fully or partially completed by a third party, either paid or unpaid
- using output from artificial intelligence tools (e.g. ChatGPT) in whole or part without acknowledgement and/or without the approval of the Unit Convenor
- using another person's work or ideas as though it is your own work, without appropriate attribution
- working closely with another student or group of students (either past or current), to submit for assessment, some or all of the other student or students' work as your own work
- sharing without permission of the Unit Convenor, Swinburne resources or other material related to assessment to an entity or document repository site
- creating, intentionally modifying or inventing information that is intended to be submitted as part of an assessment item
- using the whole or part of a computer program written by another person as your own without appropriate acknowledgement
- poorly paraphrasing somebody else's work
- using a musical composition or audio, visual, graphic and photographic work created by another person without acknowledgment
- enabling others to cheat, including letting another student copy your work or by giving access to a draft or completed assignment
- letting someone or something else impersonate you, or you impersonate someone else in an invigilated or non-invigilated assessment item
- accessing, obtaining and/or providing to others unauthorised materials relating to an invigilated or non-invigilated assessment item.

The penalties for academic misconduct can be severe, ranging from a zero grade for an assessment task through to exclusion from Swinburne. For further details, see <https://www.swinburne.edu.au/student-login/academic-integrity/>

Student support

Swinburne offers a range of services and resources to help you complete your studies successfully, such as study and learning support and health and wellbeing services. Your Unit Convenor or studentHQ can provide information about the study support and other services available for Swinburne students. For further information, see the [Student support services](#) web page.

Special consideration

If your studies have been adversely affected due to serious and unavoidable circumstances outside of your control (e.g. severe illness or unavoidable obligation), you may be able to apply for special consideration (SPC).

Applications for Special Consideration are submitted via the SPC online tool normally no later than 5.00pm on the third working day after the submission/sitting date for the relevant assessment component. See <https://www.swinburne.edu.au/life-at-swinburne/student-support-services/special-consideration-assistance/>

Note: Submitting fraudulent (fake or altered) medical certificates is considered misconduct and can lead to serious penalties from Swinburne. In addition, your doctor may report fraudulent medical certificates as a prosecutable offence under the Victorian Crimes Act.

AccessAbility Services

If you are a student with a disability, medical or mental health condition or you have significant carer responsibilities, you may require reasonable adjustments to fully access and participate in education. Swinburne's AccessAbility Services can develop an Education Access Plan (EAP) that includes the services and reasonable adjustments that you need.

It is recommended that you register with AccessAbility Services when you first commence your course, but you can contact the service at any time during your studies to find out about reasonable adjustments. Contact [Accessability Services](#) to discuss further.

Once you have an approved EAP, you should contact the Unit Convenor as early as possible (e.g. beginning of semester) to discuss your adjustments so that we can better support your learning.

Review of marks

An independent marker reviews all fail grades for major assessment tasks. In addition, a review of assessment is undertaken if your final result is between 45 and 49 or within 2 marks of any grade threshold.

You can ask the Unit Convenor to check the result for an assessment item or your final result. Your request must be made in writing within 10 working days of receiving the result. The Unit Convenor can discuss the marking criteria with you and check the aggregate marks of assessment components to identify if an error has been made. This is known as local resolution. If you are dissatisfied with the outcome of the local resolution, you can lodge a formal complaint.

Feedback, complaints and suggestions

In the first instance, discuss any issues with your Unit Convenor. If your concerns are not resolved or you would prefer not to deal with your Unit Convenor, then you can complete a feedback form. See <https://www.swinburne.edu.au/corporate/feedback/>

Advocacy

If you require advice and support navigating University regulations, policies and procedures, or responding to academic progress, misconduct or assessment matters, you are advised to seek advice from an Advocacy Officer at the Swinburne Student Association. Talking to an Advocacy Officer is free, independent and confidential. For more information and booking an appointment, please see [Advocacy at the Swinburne Student Association](#)